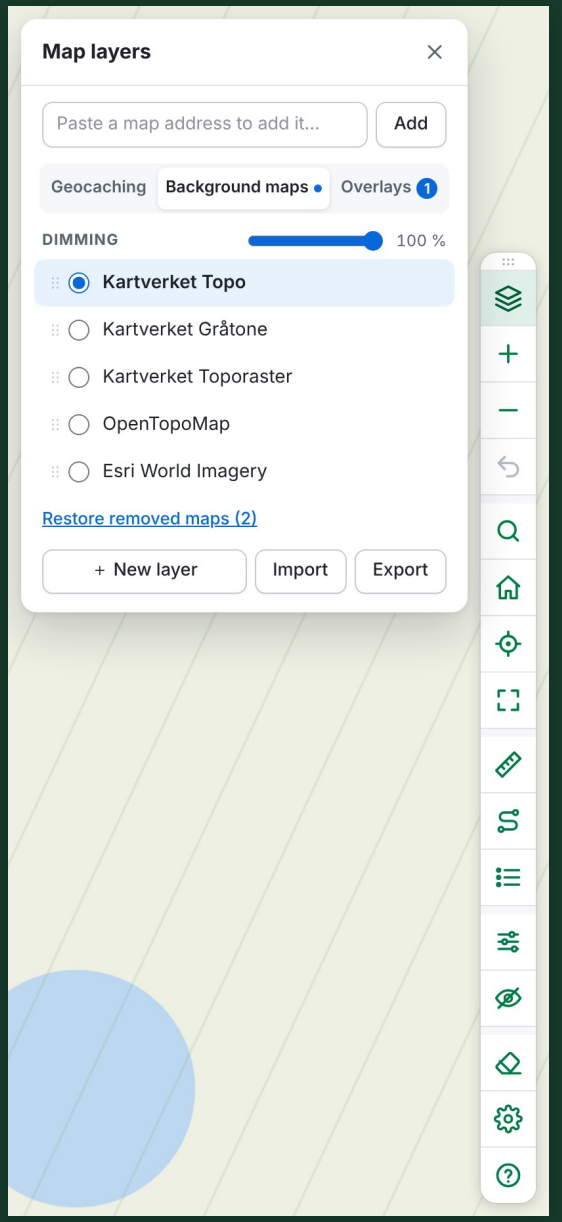
Geocaching Map – Custom Tile Layers

Your own maps and handy tools right in the geocaching.com maps

Userscript · version 1.55.32 · English and Norwegian



Everything you need on the map, in one toolbar

Any map

Kartverket, OpenStreetMap, aerial photos – or your own XYZ, WMTS, WMS and ArcGIS services

Place search

Place names and addresses from Kartverket, places abroad and coordinates – in one list

Coordinates and projecting

DMM, DD, DMS and UTM, projection and measuring distance and bearing

161 m circles

Around traditional and solved caches and visible physical stages – with clickable marks

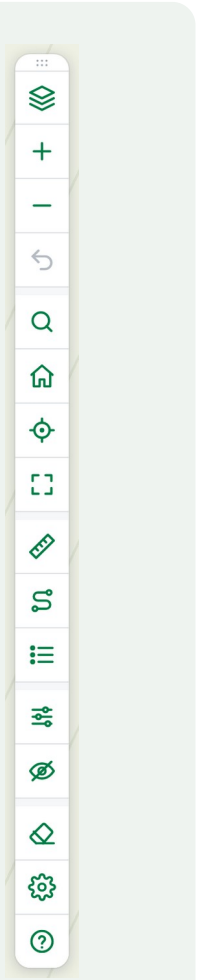
Home, location, Street View

Go home or to your location, and open Street View from the right-click menu

New: routes for your GPS

Pick caches and pins from a list or on the map, preview and download as a GPX route

One toolbar instead of many buttons



Map layers
Choose background map and overlays

Previous view
Back one step at a time – up to 10

Find a place
Shortcut: press / on the map

Home
Right-click to change the address

My location
With an accuracy circle

Fullscreen
The map fills the whole screen

Measure
Distance and bearing – snaps to caches

Route
Build a route and download it as GPX

Geocaches on the map
A list that follows the map (/map)

Show on map
Circles, marks, favourites, D/T, FTF and type icons

Hide geocaches
A clean map – click again to show them

Clear drawings
Marks, measurements and circles

Settings
Layout, language and behaviour

Help
A one-page overview, with a link to the guide

Show on map – everything in one place

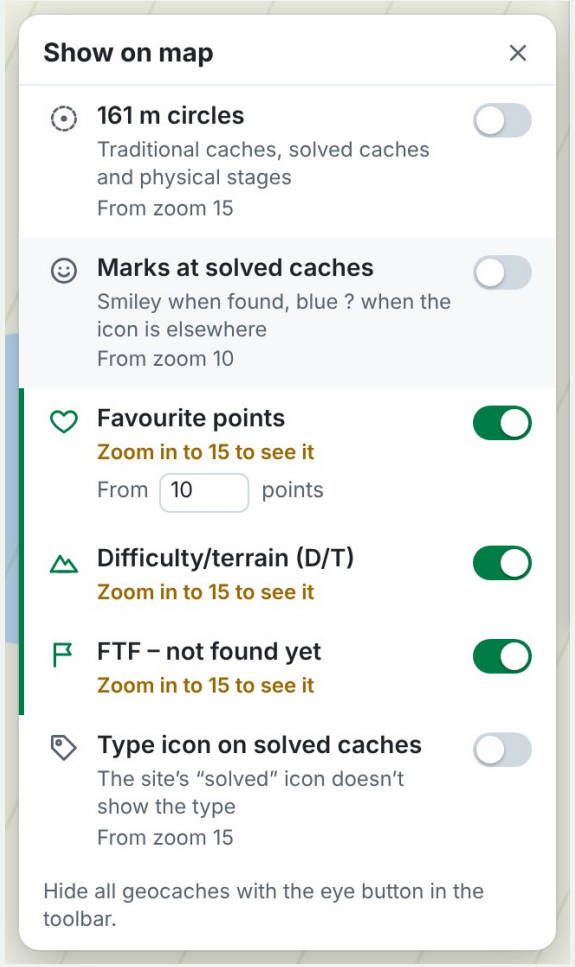
161 m circles
Traditional and solved caches and physical stages – from zoom 15

Marks at solved caches
Smiley when found, blue ? when the icon is elsewhere – from zoom 10

Favourite points and D/T
The number above the caches, difficulty/terrain below – from zoom 15

FTF and type icon
Not found yet, and the cache type on solved caches – from zoom 15

Help and press and hold
A one-page overview; press and hold gives the right-click menu on an iPad



Choose maps and overlays

Three tabs

Geocaching.com's own maps, background maps and overlays – each on its own, so the list stays short

Background and overlays

Kartverket, OpenTopoMap, OpenStreetMap and Esri – plus your own. As many overlays as you like

Fading towards white

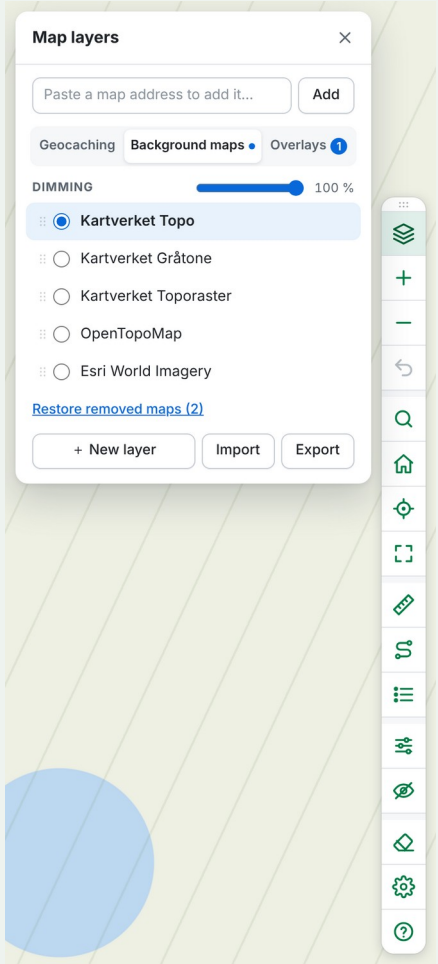
The background can be faded by up to 50 %, so the caches stand out

Your order

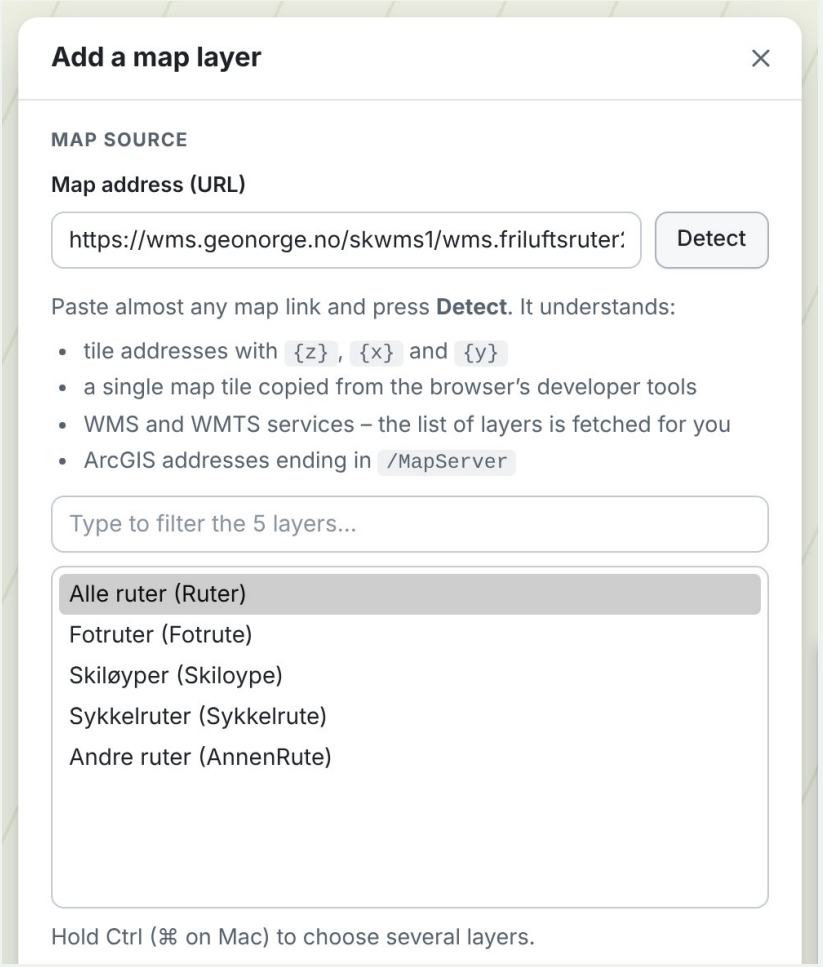
Drag the maps up and down. The overlay at the top of the list is drawn on top

Tidy the list

Remove maps you don't use – also geocaching.com's own – and bring them back whenever you like



Add your own maps



Paste almost any map link and press Detect:

XYZ / TMS
Tile addresses with {z} {x} {y}

WMTS
The list of layers is fetched automatically

WMS
Pick one or more layers

ArcGIS
Addresses ending in /MapServer

- Preview on the map before you save
- Import and export your layers as a file
- A single map tile from the developer tools is enough

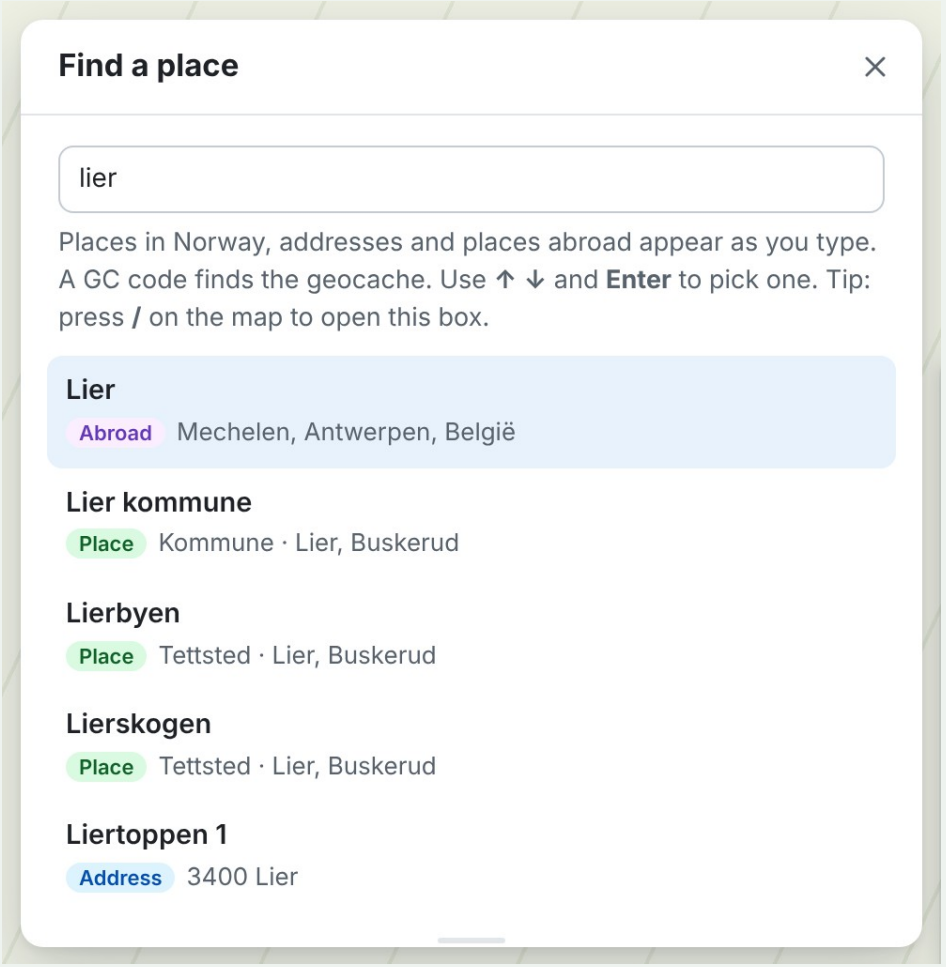
Find places, addresses and coordinates

One combined list
Hits appear as you type, best match first – whatever the source

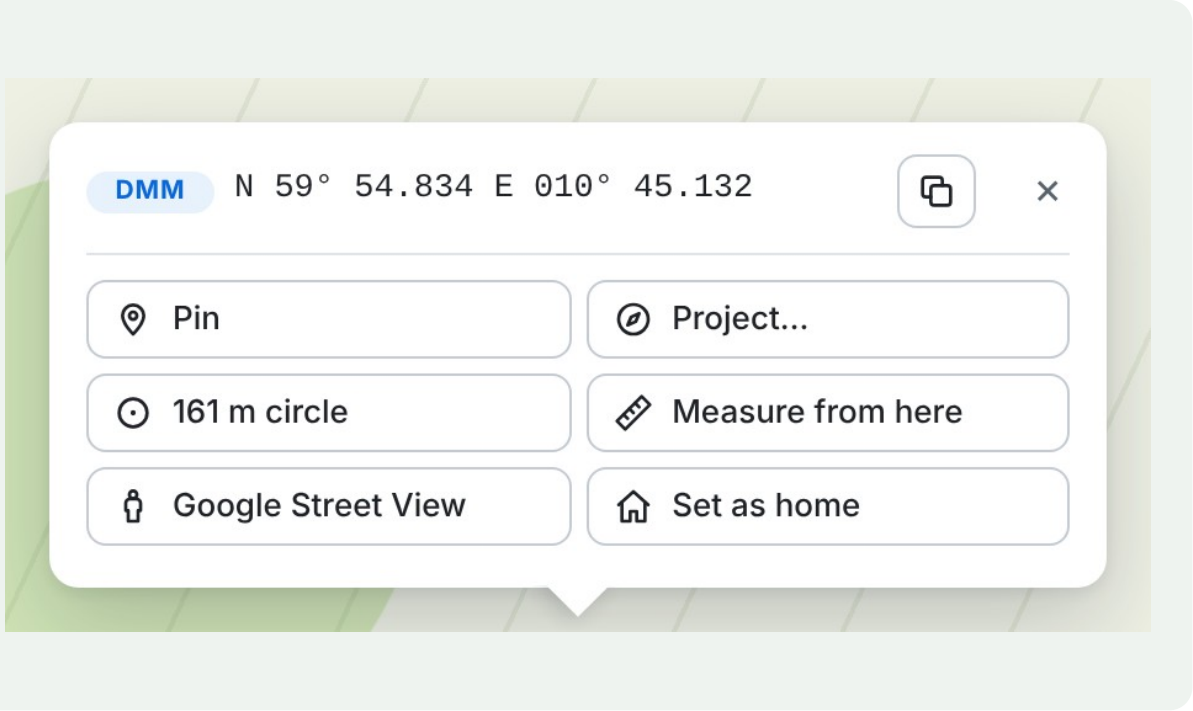
Places, addresses and abroad
Kartverket's place names and addresses, and places abroad via Photon (OpenStreetMap)

Coordinates
Type DMM, DD, DMS or UTM – the script recognises them

Home address
The same search box is used to save your home address



Coordinates in every format



Format	Example (Oslo)
DMM	`N 59° 54.834 E 010° 45.132`
DD	`59.913900, 10.752200`
DMS	`N 59° 54' 50.0" E 10° 45' 07.9"`
UTM	`32V 597980 6643119`

Right-click the map: one coordinate line that opens in DMM. Click the line to switch format, and copy with one click. The Google Street View button opens Street View at the spot. The zoom level is at the bottom – type a number to zoom.

UTM in the point's own zone, as zone, easting and northing. Accurate to about a millimetre (Krüger series).

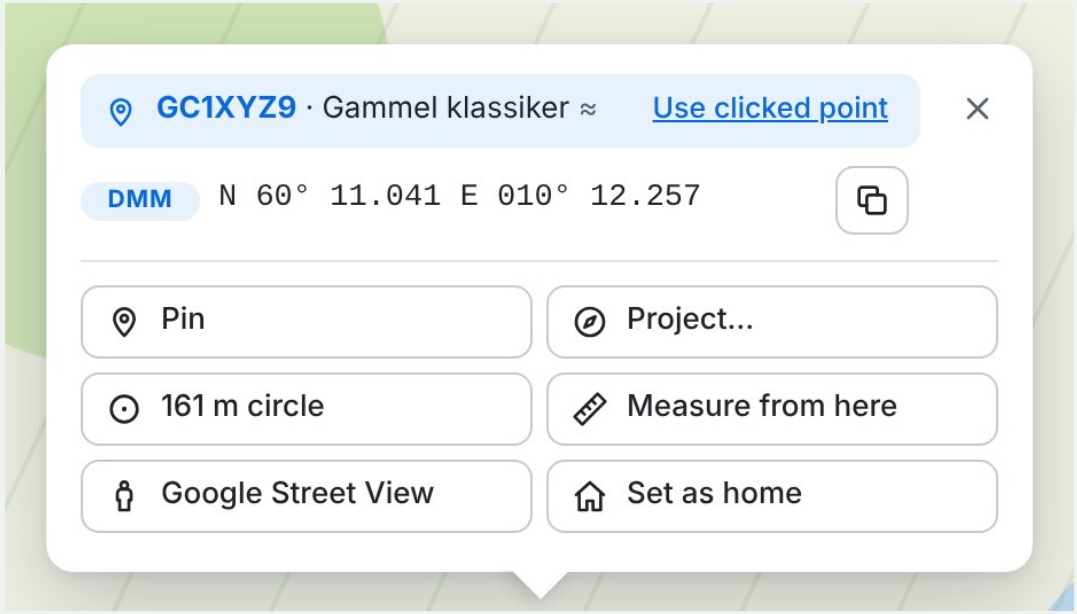
Right-click a cache – straight to its coordinates

Snaps to the cache
Right-click on or right next to a geocache, and the popup uses the cache's coordinates

GC code, name and ❤️
Favourite points are shown; click the GC code to open the cache page

All tools come along
Pin, projecting, measuring and the 161 m circle start from the cache

Use the clicked spot
One click switches to the exact spot you clicked



Exact coordinates from geocaching.com's cache search – also on /map. Only before the search has answered is the position worked out from the icon (≈).

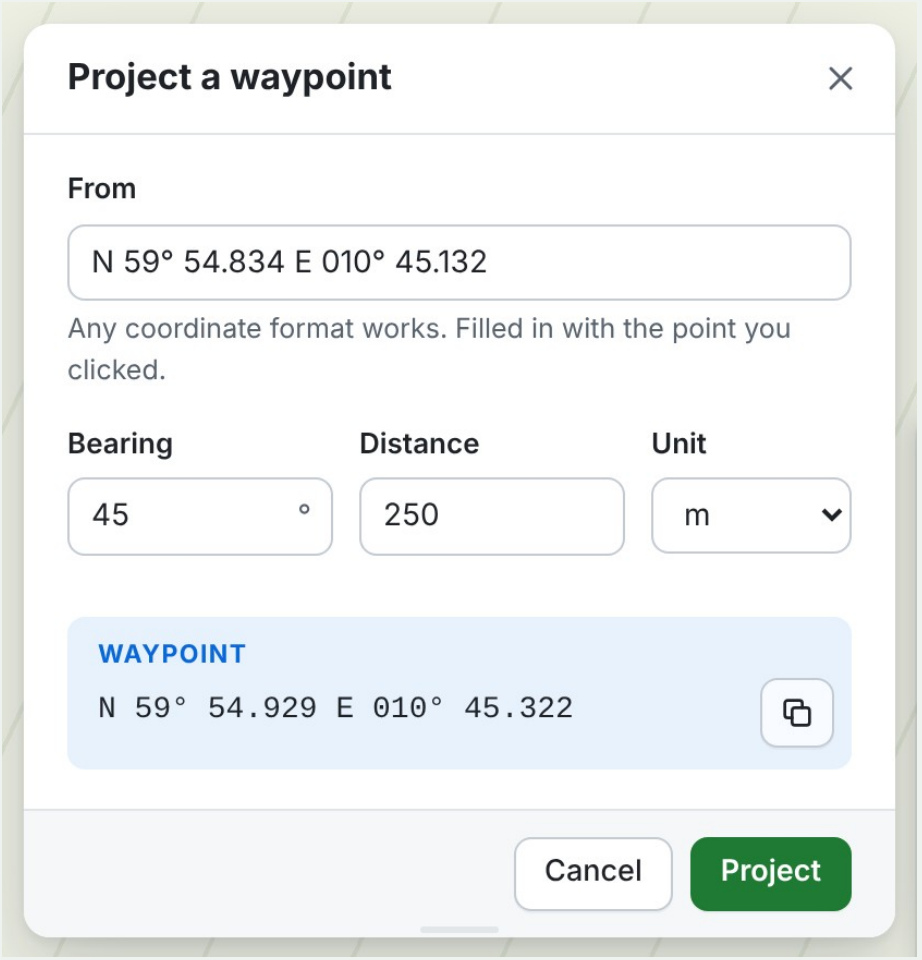
Project waypoints and measure distances

Project from a point
The start point is filled in for you. Type bearing and distance in m, km or ft

See the result at once
The waypoint shows as you type, with a dashed line on the map and a copy button

Measure distance and bearing
Click points on the map – total length and the last leg are shown

Good enough for geocaching
The error is well below a metre at ordinary distances



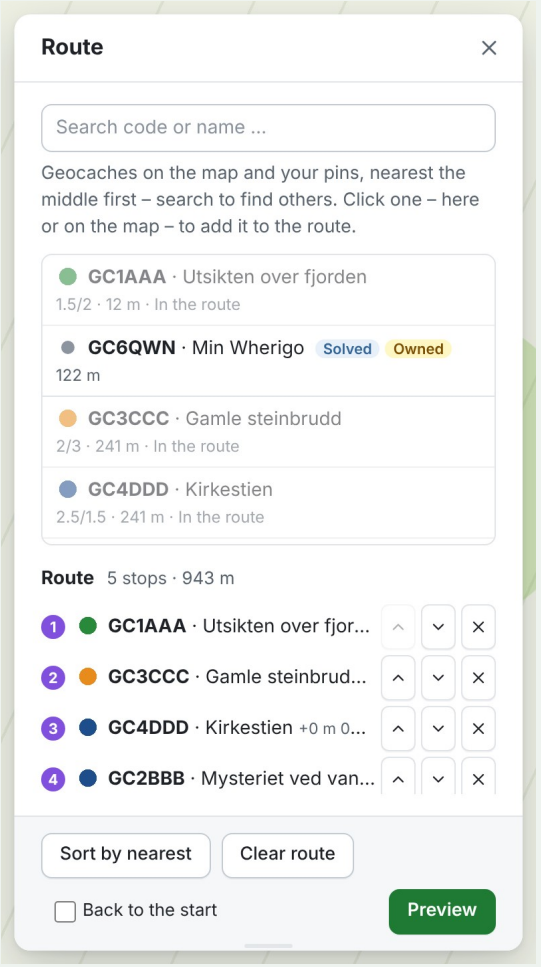
New: build a route and take it with you on your GPS

Searchable list
Nearby geocaches and pins – and any GC code worldwide

Pick on the map
Click caches and pins while the route window is open, or use the right-click popup

Order and round trip
Move stops, sort by nearest, see every leg – and end at the start

Preview and download
GPX with Garmin symbols – optionally with full cache data (paperless caches)



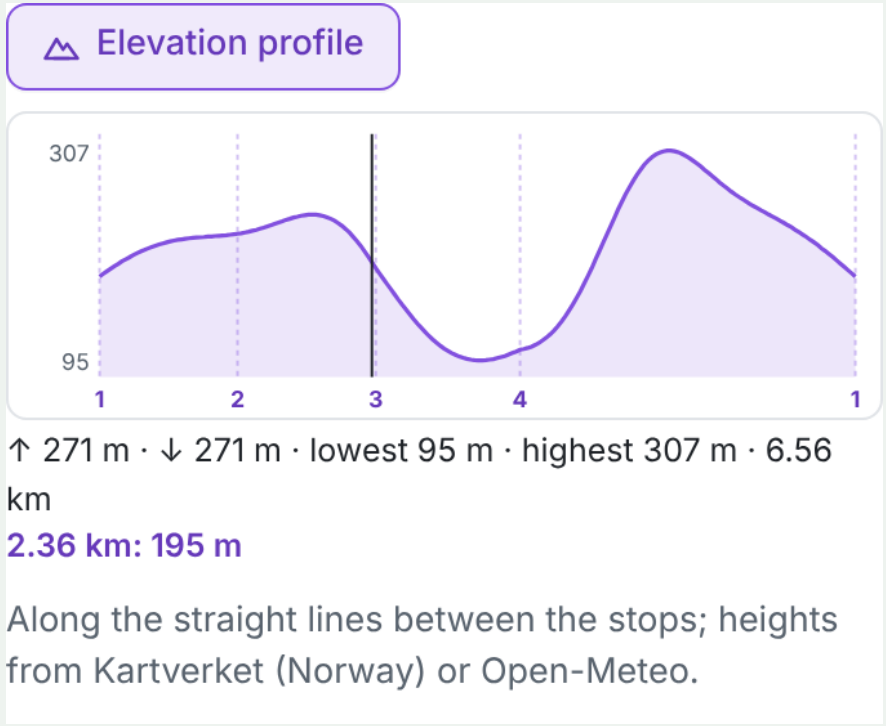
New: elevation profile for the route

Heights along the whole route
With the stops marked – also the way back on a round trip

Climb and descent
Total climb and descent, the lowest and highest point, and the length

Point and see
The mouse along the profile shows distance and height – and the spot on the map

Kartverket and Open-Meteo
The national elevation model in Norway, Copernicus 90 m elsewhere



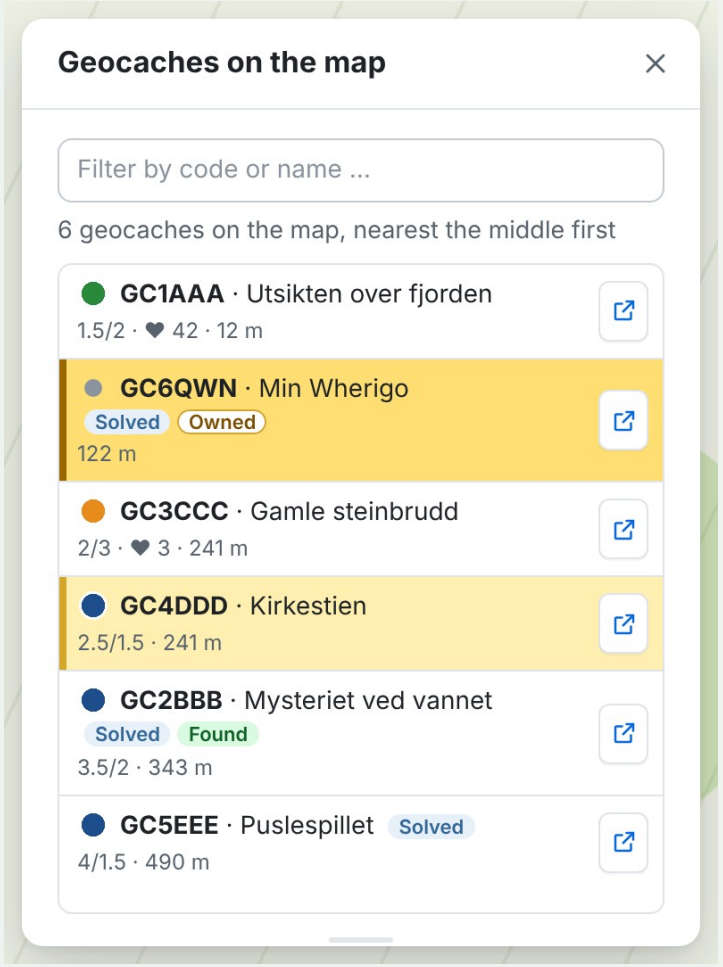
New: the geocaches on the map in a list

Follows the map
The geocaches on the map you see, nearest the middle first – with Solved, Found, Owned, D/T and ❤️

Point and find
Point at a cache on the map and it is highlighted in the list. Filter by code or name

Zoom with a glow
Click to zoom all the way in – the cache glows on the map. Click again to go back

The cache page
The button on the line opens the cache page in a new tab



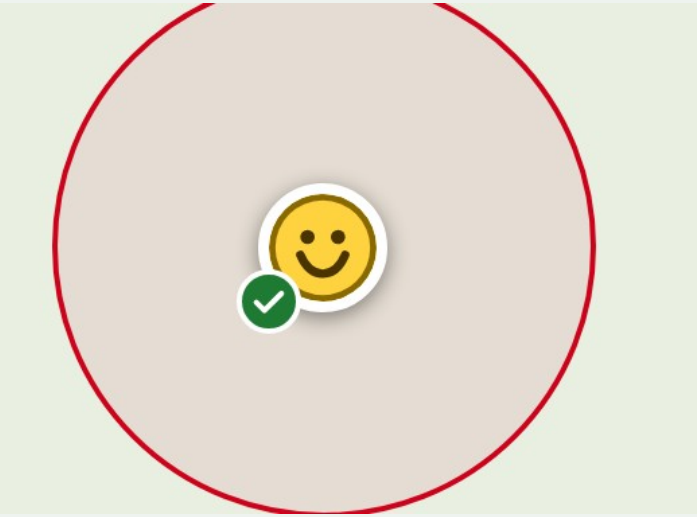
161 m circles where something is placed

Traditional caches
At their exact posted coordinates

Solved caches
At your corrected coordinates – also when the icon has moved back after a find; a small type icon can show the cache type

Physical stages and finals
Visible stages of multis, mysteries, Wherigos and letterboxes, read from the cache page

Single circle
Right-click anywhere and choose 161 m circle



Where no cache icon shows the spot, a clickable mark sits in the middle of the circle. Hovering over a solved cache shows “Final location”; right-click gives the exact coordinates.

161 m circles, cache pages and home



161 m circles

- Where something is placed – see the previous slide
- Shown from zoom level 15
- Follow the map and never block clicks
- Single circle: right-click



Cache pages

- The small map shows your chosen map
- Same fading and overlays
- Keeps the site's view
- Never changes your choice



Home

- One click takes you home
- The first time you look up the address
- Stored only in your browser

How the map layers are stacked

ABOVE THE MAP TILES

Controls	`controls · 1000`	the toolbar, the layer panel, the coordinate display and the credits
Popups	`popupPane · 700`	the right-click popup and other map popups
Markers	`markerPane · 600`	the cache icons on /play/map, pins, labels and your location
Drawings	`overlayPane · 400`	161 m circles, measuring lines, projection lines and the accuracy circle

THE MAP TILES (TILEPANE · 200) – SORTED BY Z-INDEX

The site's other tiles	`z = bz + 10`	the cache icons on /map – always lifted above your layers
Preview	`z = bz + 3`	the layer you are trying in the layer editor
Your overlays	`z = bz + 2`	hiking routes, property boundaries, hillshade – their own opacity slider
Your background map	`z = bz + 1`	fully opaque – faded towards white with a colour filter, at most 50 %
The geocaching.com map	`z = bz`	the site's background – covered by your map, faded when it is chosen itself

bz = the z-index of the site's background. On /play/map the site's background is a vector map (WebGL). On /map an invisible cache grid (UTFGrid) is loaded as well, used for the 161 m circles and for snapping the right-click popup to caches.

Under the hood

Never removes the site's layers
The script slots its own layers in between, sorted by z-index

Finds the map by itself
Catches Leaflet as the page loads, also on /play/map and the cache pages

No seams between the tiles
Tiles are drawn 1 pixel larger, also at 110 and 125 % display scaling

Leaves the form alone
No button can submit the page's ASP.NET form, so the page never reloads

Fading without show-through
A colour filter bleaches the map towards white – the site's map never shows through

Built-in troubleshooting
gcctReport(), gcctCacheDebug(), gcctStageTest() and gcctSnapTest() in the console give short reports without personal data

Get started in three minutes

1

Install a userscript manager

Tampermonkey or Violentmonkey in Chrome, Edge, Firefox or Safari

2

Add the script

Open geocaching-custom-tiles.user.js and save it in the manager

3

Open a map

geocaching.com/map or /play/map – the toolbar appears on the right

Privacy

The script collects and sends no personal data and has no server of its own. It only contacts the map servers you switch on, Kartverket and Photon when you search, and geocaching.com's own cache search and cache pages when the 161 m circles are on. Your location is only requested when you ask for it. All settings are stored in the browser.

Better maps for better trips

User guides in English and Norwegian are included. Use gcctReport() in the browser console if something doesn't work.

